

Mason Bane

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Summary: Recent Computer Science and Engineering graduate from Tarleton State University with experience across embedded firmware, high-performance simulations, and Linux systems. Builds secure STM32-based telemetry, optimizes CUDA/C++ models for research, and maintains reliable HPC lab environments through automation and documentation. Combines hands-on hardware troubleshooting with clear communication to support research, production, and infrastructure teams.

EDUCATION

Tarleton State University

Bachelor of Science in Computer Science, Concentration in Computer Engineering

Minor in Mathematics

GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Stephenville, TX

Aug. 2022 – May 2026

Hill College

Associate of Arts in Liberal Arts

Hillsboro, TX

Aug. 2019 – Sept. 2023

EXPERIENCE

Undergraduate Research Assistant - Lead Programmer

May 2024 – May 2026

Tarleton State University

Stephenville, TX

- Developed and optimized various N-body simulations utilizing C/C++, enhancing computational efficiency and accuracy through advanced algorithms and parallel processing with CUDA.
- Collaborated with interdisciplinary teams to create digital twins to model complex problems using OpenGL and Blender, translating scientific concepts into interactive simulations and improving data interpretation.
- Conducted rigorous testing and debugging of simulation software, ensuring robust performance and reliability while documenting processes to facilitate knowledge transfer and future research initiatives.

Undergraduate Technology Specialist - HPC Lab Manager

Aug. 2024 – May 2026

Tarleton State University

Stephenville, TX

- Managed 15+ Linux systems in a production-like research environment, ensuring 99%+ uptime through proactive maintenance, hardware diagnostics, and rapid incident response
- Diagnosed and resolved complex hardware/software issues including memory failures, storage degradation, and OS configuration problems, minimizing system downtime and maintaining optimal performance
- Developed and maintained comprehensive documentation for system configurations, troubleshooting procedures, and lab protocols, enabling knowledge sharing and process standardization

Undergraduate Lab Instructor, Texas Government

January 2023 – December 2023

Tarleton State University

Stephenville, TX

- Guided and supported students in policy research and analysis, facilitating the exploration of bills and regulations to enhance critical thinking and analytical skills.
- Conducted regular meetings and collaborated with the lecture professor and fellow lab leaders to align objectives and improve instructional methodologies, ensuring a cohesive educational experience.

PROJECTS

N-body Digital Twin of the Left Atrium | NIH Grant #1R15HL179671-01 | C, CUDA Aug. 2024 – May 2026

- Engineered a parallel N-body model of the left atrium using CUDA, with a 20,000+ node mesh to simulate atrial arrhythmias in near real-time.
- Developed an intuitive C++/ImGui interface to control simulation parameters and visualize outputs, bridging the gap between complex computational models and end-user (research/clinical) requirements.
- Presented findings at academic conferences, demonstrating the tool's potential to improve clinical decision-making and transform training for medical professionals.

Secure IoT Sensor Node with Hardware TrustZone | C, STM32CubeIDE/HAL, I2C/SPIDec. 2025 – May 2026

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.

- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

N-Body Simulation of Microplastic Coagulation | *C/C++, CUDA, Linux, OpenGL* May 2024 – Aug. 2024

- Contributed to the development and optimization a CUDA-based N-body simulation of micro-plastic removal from water using plant polymers – leveraging NVIDIA’s CUDA technology for parallel processing and OpenGL for real-time 3D visualization, enhancing computational efficiency and user experience.
- Crafted a user-friendly interface for toggling simulation features, providing enhanced visual cues and aiding in the understanding of complex phenomena.
- Collaborated with a multidisciplinary team to define requirements, design features, and refine the simulation tool, ensuring alignment with project goals and user needs.

Analog Pink Noise Generator Circuit | *LTSpice, Python, MATLAB, Breadboarding* Aug. 2025 – Dec. 2025

- Designed and prototyped a BJT-based pink noise generator with active filter stage using LTSpice for simulation and breadboarding for physical implementation.
- Developed Python scripts to quantitatively compare simulation results with an ideal pink noise spectrum, guiding component selection and filter tuning for improved accuracy.
- Built a physical demonstration circuit with audio output, successfully verifying simulation models against real-world performance.
- Collaborated in a two-person team to present comprehensive design methodology, results, and data-driven decision process to peers and professors.

PUBLICATIONS

• **Near Real-Time Interactive N-body Model of the Left Atrium for Exploratory Arrhythmia Induction and Ablation Simulation**

Journal: Heart Rhythm O2, June 2026

Authors: Bryant Wyatt, Leah Rogers, Mason Bane, Kyla Moore, Philip Alcorn, Gabriella Williams, Kinsey Brawner, Madhur Wyatt, Charles Puelz

Our team developed a real-time interactive N-body model of the left atrium for exploratory arrhythmia induction and ablation simulation.

[Read Paper](#)

• **Investigating Atrial Fibrillation Mechanisms Using an Interactive N-body Digital Twin of the Left Atrium**

Abstract: Heart Rhythm 23(4):S136, April 2026

Authors: Leah Rogers, Kyla Moore, Mason Bane, Philip Alcorn, Gabriella Williams, Kinsey Brawner, Charles Puelz, Bryant Wyatt, Madhur Wyatt

Our team’s interactive computational model investigates mechanisms of atrial fibrillation and supports evaluation of therapeutic strategies.

[Read Abstract](#)

• **Interactive Digital Twin of the Left Atrium for Simulating Atrial Arrhythmias and Virtual Catheter Ablation**

Abstract: Heart Rhythm 23(4):S130, April 2026

Authors: Bryant Wyatt, Mason Bane, Leah Rogers, Kyla Moore, Gabriella Williams, Kinsey Brawner, Philip Alcorn, Charles Puelz, Madhur Wyatt

Our interactive digital twin simulates atrial arrhythmias and supports virtual catheter ablation for clinical training and procedural assessment.

[Read Abstract](#)

- **"Offloading" Undergraduate Research to the Graphics Processing Unit for Acceleration**
Conference: SC Workshops '25, Nov 2025
 Authors: Bryant Wyatt, Mason Bane
 We demonstrate methods for accelerating undergraduate research computations using GPU parallel processing (CUDA), achieving substantial performance improvements on scientific workloads.
[Read Paper](#)
- **Studying Left Atrial Arrhythmias Using a Real-Time Interactive Digital Twin**
Journal: Heart Rhythm O2 6(9):S2, Sept 2025
 Authors: Bryant Wyatt, Gavin McIntosh, Avery Campbell, Milanie Little, Brandon Wyatt, Mason Bane, Leah Rogers, Kyla Moore, Conner Homrighaus, Charles Puelz
 Our group developed a real-time digital twin of the human left atrium used to study atrial arrhythmias and support catheter ablation planning and medical education.
[Read Paper](#)

AWARDS & RECOGNITION

- **Magna Cum Laude**, Tarleton State University, May 2026
- **First Place Undergraduate Poster**, Mathematical Association of America – Texas Section, Mar. 2026
- **First Place Undergraduate & Graduate Poster**, Tarleton State University REID Research Symposium, Feb. 2025
- **President's List** (Fall 2022, Spring 2023, Fall 2023)
- **Dean's List** (Spring 2025, Fall 2025)
- **Accepted Poster**, NVIDIA GTC 2026

CERTIFICATIONS

- **30-Hour General Industry Safety and Health**, Issued 2021, ID: 25-900506266

PROFESSIONAL MEMBERSHIPS

- SIAM (Society for Industrial and Applied Mathematics)
- SAMPE (Society for the Advancement of Material and Process Engineering)
- IEEE (Institute of Electrical and Electronics Engineers)
- AIAA (American Institute of Aeronautics and Astronautics)
- Tarleton Computer Society
- Tarleton Drone Team

TECHNICAL SKILLS

Microcontroller Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi, ARM Cortex-M
Firmware & Languages: C, C++, ARM Assembly, Python, Java, Bash, HTML/CSS/JavaScript, I²C, SPI, UART
Hardware Tools: STM32CubeIDE, LTSpice, PCB Prototyping, Oscilloscope, Multimeter, GDB
Development & Systems: Git, Linux, CMake/Make, Visual Studio, VS Code, RTOS Concepts
Research & Simulation: MATLAB, CUDA, OpenGL, Blender, ImGui

References available upon request